

Quiz (Habanero)

- Q1.** Factor the expression $x^2 - 9$:
(A) $x^2 + 9$
(B) $(x + 3)(x - 3)$
(C) $(x - 3)(x + 3)$
(D) $x^2 - 3$
(E) $x^2 + 3$
- Q2.** The area of a drought-affected region is $a^2 - 16$. Factor the expression to find the length and width:
(A) $(a - 4)(a - 4)$
(B) $(a + 4)(a - 4)$
(C) $(a - 4)(a + 4)$
(D) $(a - 2)(a - 8)$
(E) $(a - 8)(a - 2)$
- Q3.** Factor the expression $49 - y^2$:
(A) $(y - 7)(y + 7)$
(B) $(y + 7)(y - 7)$
(C) $(y - 7)(y - 7)$
(D) $y^2 - 49$
(E) $y^2 + 49$
- Q4.** A hurricane's wind speed can be modeled by $x^2 - 25$. Factor the expression to determine the possible wind speeds:
(A) $x^2 + 25$
(B) $(x + 5)(x - 5)$
(C) $(x - 5)(x + 5)$
(D) $(x - 25)(x + 25)$
(E) $x^2 - 5$
- Q5.** Factor the expression $64 - z^2$:
(A) $(z - 8)(z + 8)$
(B) $(z + 8)(z - 8)$
(C) $(z - 8)(z - 8)$
(D) $z^2 - 64$
(E) $z^2 + 64$
- Q6.** A researcher is studying the temperature changes in a forest ecosystem. The temperature can be modeled by $t^2 - 36$. Factor the expression:
(A) $t^2 + 36$
(B) $(t + 6)(t - 6)$
(C) $(t - 6)(t + 6)$
(D) $t^2 - 6$
(E) $t^2 + 6$
- Q7.** The number of endangered species in a region can be modeled by $x^2 - 49$. Factor the expression to determine the possible number of species:
(A) $x^2 + 49$
(B) $(x + 7)(x - 7)$
(C) $(x - 7)(x + 7)$
(D) $x^2 - 7$
(E) $x^2 + 7$
- Q8.** The altitude of a mountain can be modeled by $a^2 - 100$. Factor the expression:
(A) $a^2 + 100$
(B) $(a + 10)(a - 10)$
(C) $(a - 10)(a + 10)$
(D) $a^2 - 10$
(E) $a^2 + 10$

Open Ended Questions (FRQ)

- Q9.** A city's rainfall can be modeled by the expression $x^2 - 121$. Factor the expression and use it to determine the possible values of x if the rainfall is 0. Provide your answer in the space below.
- Q10.** A scientist is studying the water levels in a lake. The water level can be modeled by the expression $y^2 - 225$. Factor the expression and use it to determine the possible values of y if the water level is 0. Provide your answer in the space below.

Chef's Tips

- Tip 1: Encourage students to use factoring to simplify expressions and solve problems.
- Tip 2: Remind students that the difference of squares pattern can be applied to various real-world scenarios, including weather and environment.
- Tip 3: Emphasize the importance of checking answers by plugging them back into the original equation.